MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL,
QUADRUPLE, 2 INPUT OR GATES,
MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/335B, dated 24 October 1984, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete and substitute as printed above.

1.2.3, add the following case outline:

"X C-2A (20-terminal, .350" x .350"), square chip carrier package"

1.3, Thermal resistance, delete the following in its entirety:

"Cases (all LCC'S) - - - - - - - - - 0.08°C/W 2/"

Footnote 2/: Delete in its entirety.

PAGE 2

3.2.2, delete and substitute the following:

"3.2.2 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request."

PAGE 3

TABLE I, Low-level input current, Unit column: Delete "µA" and substitute "mA".

PAGE 4

TABLE II, Interim electrical parameters: Delete "(pre burn-in)".

TABLE II, Group A test requirements, Class B devices: Add "10,11".

TABLE II, Additional electrical subgroups for group C periodic inspections: Delete in its entirety.

4.2a(1), following test condition D: Add "E, and F".

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
7	7
8	Reprinted without change

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PAGE 5

4.4.3b: Delete in its entirety.

4.4.3c(1), following test condition D: Add "E, and F".

PAGE 9

TABLE III, Device type 01: Delete "Case 2" and substitute "Cases 2, X".

PAGE 10

TABLE III, Device type 01: Delete "Case 2" and substitute "Cases 2, X".

Footnote 1/, delete and substitute the following:

"1/ For cases 2 and X, pins not referenced are N/C."

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

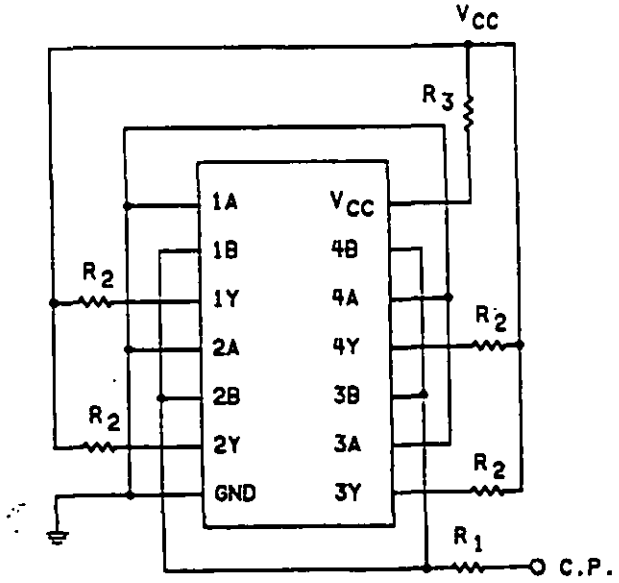
(Project 5962-1094)

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Device type 01.

Cases A, B, C, D, X, 2

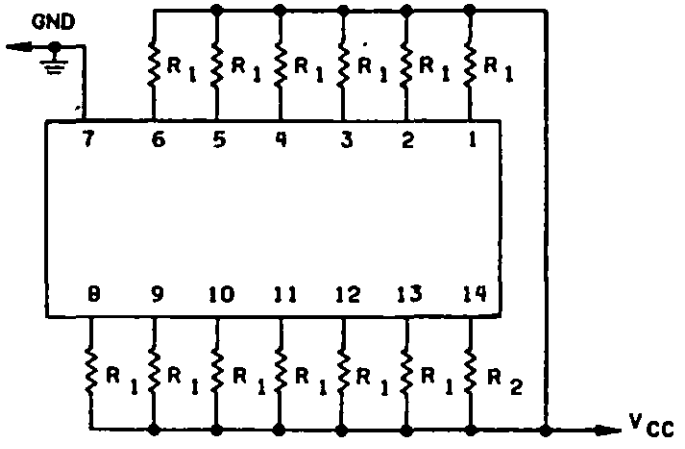
(pins not referenced for cases X and 2 are N/C)



NOTES:
1. CP = 100 kHz ±50 percent square wave; duty cycle = 50 ±15 percent; V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R_1 = 160Ω ±5 percent, R_2 = 270Ω ±5 percent.
3. V_{CC} and R_3 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

Device type 01.

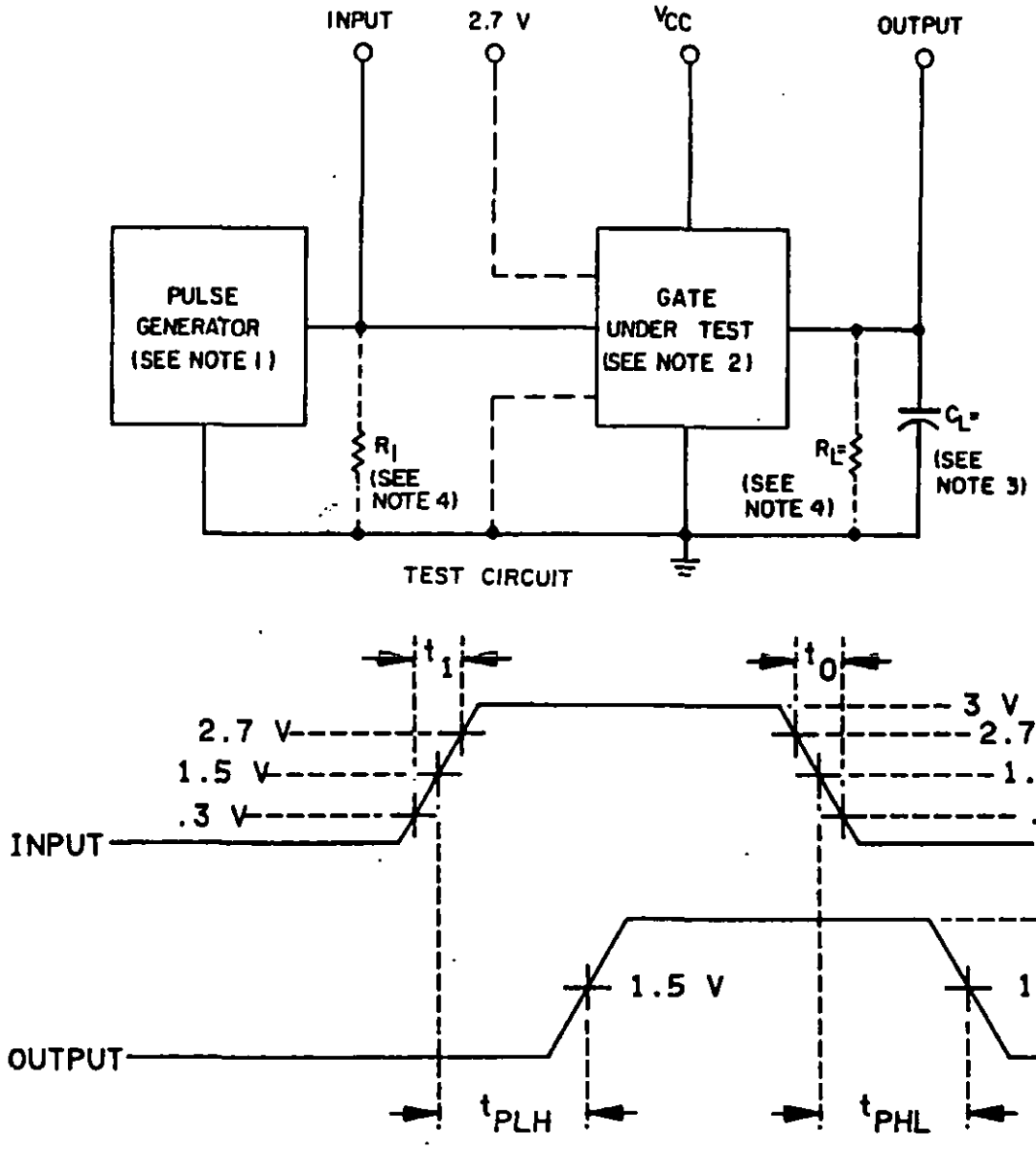
Case A, B, C, and D



NOTES:
1. High temperature condition F (175°C) burn-in circuit only.
2. R_2 = 2Ω.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at V_{CC} terminal.

FIGURE 2. Burn-in and life test circuit.

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NOTES:
1. Input pulse has the following characteristics: $t_1 = t_0 \leq 2.5$ ns; PRR $\leq$ 1 MHz, $Z_{OUT} = 50\Omega$.
2. Inputs not under test are at ground.
3. C_L = 50 pF ±10 percent including scope probe, wiring and stray capacitance without package in test fixture.
4. R_L = 499Ω ±5 percent, R_1 = 50Ω ±5 percent.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 3. Switching time test circuit and waveforms.

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